Deep Learning for NLP

AI Text Detectors and Watermarking

AI Text Detectors and Watermarking

Learning goals

- Distinguish passive AI-text detection from active provenance and watermarking
- Explain stylometric, likelihood-based, and trained detector families
- Understand why detector performance depends on prompts, models, domains, editing, and author populations
- Explain the basic theory of text watermarking and its robustness–quality trade-offs

Human-Written

The programme operates on a weekly elimination process to find the best all-around baker from the contestants, who are all amateurs.

Generated

The first book I went through was The Cook's Book of New York City by Ed Mirvish. I've always loved Ed Mirvish's recipes and he's one of my favorite chefs.

TWO DIFFERENT PROBLEMS I

There are two fundamentally different ways to ask whether text is AI-generated.

Passive detection Infer authorship from statistical traces already present in the text.

$$x \longrightarrow \hat{y} \in \{\text{human}, \text{AI}\}$$

Active provenance Insert a signal during generation and later test whether it is present.

$$\text{generator} + \text{key} \longrightarrow x_{\text{marked}}$$

Passive detectors classify a moving distribution; watermarks create a signal.

▸ Wu et al., 2025, Fig. 4

WHAT DOES A DETECTOR OBSERVE? I

The detector never observes the writing process directly.

$$\underbrace{\text{model}}_{\text{base + post-training}} + \underbrace{\text{prompt}}_{\text{task + style + persona}} + \underbrace{\text{domain}}_{\text{topic + genre}} + \underbrace{\text{decoding}}_{\text{sampling}} \longrightarrow x$$

The detector sees only:

$$x \longrightarrow D(x)$$

Therefore, "AI-likeness" is confounded with style, topic, genre, prompt, and editing.

Stylometric Differences between Human and AI Text

CLASSICAL ROOTS: AUTHORSHIP ATTRIBUTION I

AI-text detection is related to authorship attribution.

Authorship attribution Which human author wrote this?

$$P(a_i \mid x)$$

AI-text detection Was this produced by a generative system?

$$P(\mathrm{AI} \mid x)$$

Classical stylometry uses stable linguistic habits:

- function words,
- character and word n-grams,
- sentence length,
- syntactic patterns,
- vocabulary richness.

▸ Stamatatos, 2009

STYLOMETRIC FEATURES I

Early machine-generated-text detection relied on surface statistics.

- function-word density,
- phrase and n-gram frequency,
- grammatical branching,
- constituent length,
- fluency and grammaticality features.

These features are interpretable, but not necessarily stable across generators.

▸ Wu et al., 2025, Sec. 5.2.1

STYLOMETRY IS NOT ENOUGH I

A feature can separate human and AI text in one dataset and fail in another.

$$P(f \mid \text{AI}, \text{domain A}) \neq P(f \mid \text{AI}, \text{domain B})$$

The real question is not whether a feature differs, but whether the difference transfers.

This motivates evaluating across:

- prompts,
- generator models,
- domains,
- writing and editing processes.

A GENERALIZATION BENCHMARK I

The benchmark combines:

- **6 prompts:** 0-shot, 3-shot, style, CoT, 1-shot CoT, self-refine,
- **7 LLMs:** Mistral, DeepSeek, Llama, Qwen variants, Solar,
- **4 domains:** abstracts, news, reviews, QA.

For every prompt–model–domain condition, a detector is fine-tuned and tested under distribution shifts.

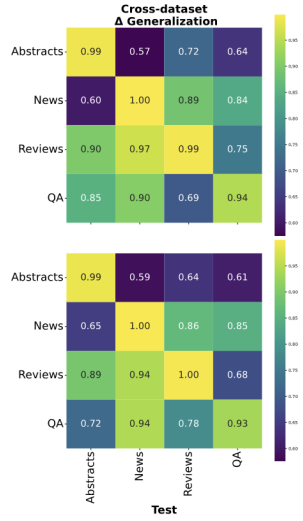
This turns detector evaluation into a controlled transfer problem.

▸ Xia, Stańczak & Roth, 2026

IN-DOMAIN SUCCESS, TRANSFER FAILURE I

Near-perfect diagonal performance does not imply robust generalization.
The accuracy of a detector can drop from 99% to 57% when the domain changes.

▸ Xia et al., 2026, Fig. 1a



FEATURE SHIFTS EXPLAIN SOME TRANSFER GAPS I

For a linguistic feature f, define the human–AI feature gap:

$$f(D) = f(D_{\text{human}}) - f(D_{\text{AI}})$$

The train–test shift is:

$$\Delta f = f(D_{\text{train}}) - f(D_{\text{test}})$$

Then correlate detector generalization difference (Δ_{gen}), the accuracy difference between train and test domain, with feature shifts:

$$\text{Corr}(f) = |\text{Pearson}(\Delta_{\text{gen}}, \Delta_f)|$$

▸ Xia et al., 2026, Eq. 4–5

ROBUST LINGUISTIC CORRELATES I

The most robust feature families include (Cross-Prompt, C-P, and Cross-Model, C-M):

- pronoun usage, verb tense, active/passive voice,
- readability and sentence structure in specific settings.

No feature was robust Cross-Domain (C-D)!

		DeBERTa-based			RoBERTa-based		
		C-P	C-M	C-D	C-P	C-M	C-D
Pearson	Bonferroni	2	15	0	3	28	0
	FDR	2	37	0	6	58	0
	Top features	Numerals Passive voice	Past tense Personal names Numerals	–	“Our” pronoun Present tense “We” pronoun	“It” pronoun Active voice “Yourself” pronoun	–
Spearman	Bonferroni	3	16	0	2	25	0
	FDR	4	44	0	10	51	0
	Top features	“She” pronoun “He” pronoun short sentences	MATTR FLESCH Gunning Fog	–	“She” pronoun “Her” pronoun Present tense	Active voice “It” pronoun Function words	–

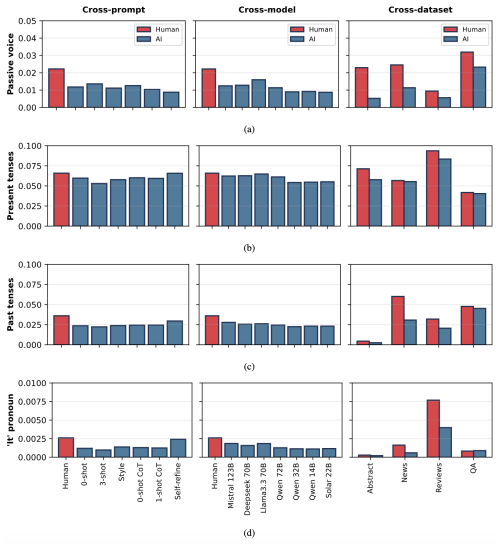
Table 3: Robustness study of applying multiple-hypothesis correction to Pearson (linear) and Spearman (non-linear) correlation. The numerical values represent the number of features that remain significant after the multiple-hypothesis correction.

▸ Xia et al., 2026, Table 3

HUMAN–AI FEATURE DIFFERENCES I

In this benchmark, AI text often uses less of some features than human text, e.g. passive voice, past tense, and the pronoun “it”.

▸ Xia et al., 2026, Fig. 4



Likelihood-Based and Unsupervised Detectors

LIKELIHOOD AND PERPLEXITY I

For an autoregressive language model M:

$$P_M(x_{1:T}) = \prod_{t=1}^{T} P_M(x_t \mid x_{<t})$$

Average log-likelihood:

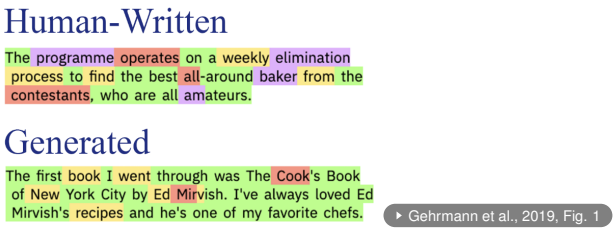
$$\frac{1}{T} \sum_{t=1}^{T} \log P_M(x_t \mid x_{<t})$$

Perplexity:

$$\mathrm{PPL}_M(x) = \exp\left(-\frac{1}{T} \sum_{t=1}^{T} \log P_M(x_t \mid x_{<t}) \right)$$

High perplexity = high average surprisal = low average log-likelihood.

GLTR: TOKEN RANKS AS VISUAL EVIDENCE I



▸ Gehrmann et al., 2019, Fig. 1

GLTR (Giant Language model Test Room) highlights each token according to its rank under a language model.

Generated text often overuses high-probability tokens.

PERPLEXITY DETECTORS USE A SCORING MODEL I

A detector can score ChatGPT text using another model, e.g. GPT-2.

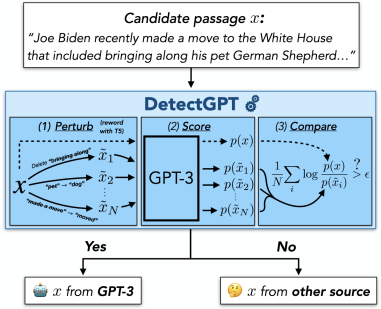
$$\mathrm{PPL}_{\text{GPT-2}}(x)$$

Useful distinction:

- generator likelihood: $P_G(x)$,
- surrogate likelihood: $P_S(x)$.

▸ Vasilatos et al., 2023

DETECTGPT: PROBABILITY CURVATURE I



DetectGPT asks whether the original text is more likely than nearby paraphrases. ▸ Mitchell et al., 2023, Fig. 1

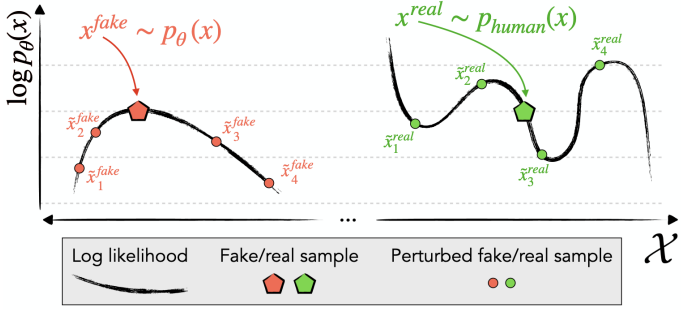
$$d(x) = \log p_\theta(x) - \mathbb{E}_{\tilde{x}\sim q(\cdot|x)} \left[\log p_\theta(\tilde{x})\right]$$

WHY “CURVATURE”? I

Machine-generated text may lie near local maxima of the model’s log-probability surface.

$$\log p_\theta(x) > \mathbb{E}_{\tilde{x}}[\log p_\theta(\tilde{x})]$$

Text is discrete, so curvature is estimated through perturbations rather than derivatives. ▸ Mitchell et al., 2023

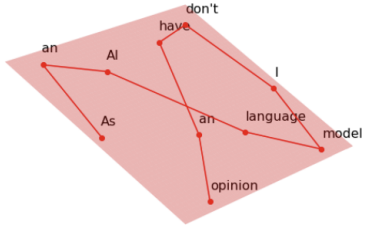


INTRINSIC DIMENSION I

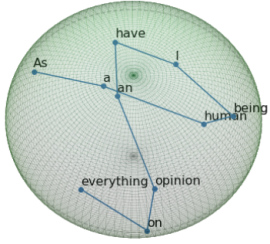
Another signal: the geometry of contextual token embeddings.

$$\{e_1, \ldots, e_T\} \subset \mathbb{R}^d$$

The text is represented as a point cloud. Its intrinsic dimension is estimated using persistent homology / MST scaling.



(a) AI generated



(b) human written

INTRINSIC DIMENSION II

Machine text may explore fewer independent representational directions.

▸ Tulchinskii et al., 2023

Trained Detectors

TRAINED DETECTORS I

Modern strong detectors are usually supervised classifiers.

$$x \xrightarrow{\text{encoder / backbone}} h(x) \xrightarrow{\text{classification head}} \hat{y}$$

RoBERTa/BERT-style detectors often perform very well in-domain, but can overfit to model and domain distributions.

In practice, the training distribution is often more important than the architecture.

▸ Wu et al., 2025, Sec. 5.3

WHERE DOES TRAINING DATA COME FROM? I

A supervised detector needs labeled examples:

$$\{(x_i, y_i)\}_{i=1}^{N}$$

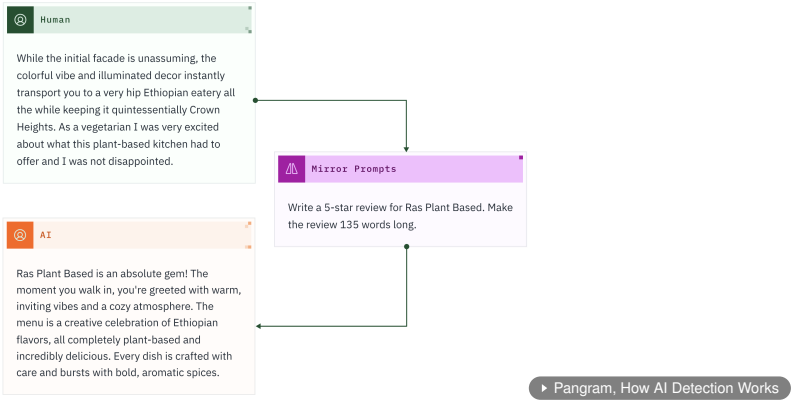
But the hard part is not just scale.

Human and AI examples must be matched for topic, genre, length, and task.

Otherwise, the classifier may learn:

news vs essays instead of human vs AI.

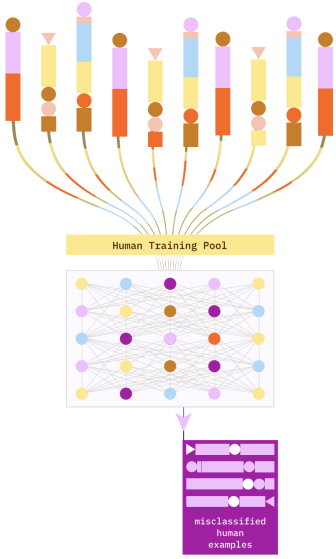
PANGRAM: SYNTHETIC MIRRORS I



Pangram describes generating AI examples that mirror human documents in topic, genre, length, and style.

$$\text{human text} \longrightarrow \text{matched synthetic mirror}$$

PANGRAM: HARD-NEGATIVE MINING I



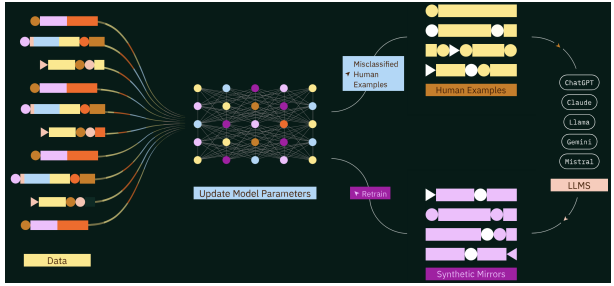
The detector is run on human corpora to find false positives.

hard human examples $\rightarrow$ training set

$\rightarrow$ retrain

▸ Pangram, How AI Detection Works

PANGRAM 4 I

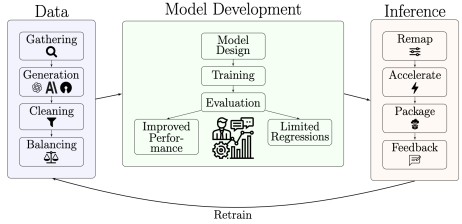


Pangram 4 reports:

- sparse mixture-of-experts backbone,
- document-level AI-assistance score,
- token-level provenance,
- mixed-authorship and humanizer heads.

▸ Pangram 4 Technical Report, 2026, Fig. 1

GPTZERO: COMMERCIAL DETECTOR PIPELINE I



GPTZero describes a supervised hierarchical detector for:

- human,
- AI,
- mixed,
- polished,
- paraphrased text.

▸ Adam et al., 2026, Fig. 1

COMMERCIAL METRICS: A CAUTION I

Vendor-reported metrics are tied to the vendor's evaluation distribution.

$$\mathrm{FPR}_{\text{vendor}} \neq \mathrm{FPR}_{\text{your population}}$$

Ask:

- Which domains and authors?
- Which LLMs and model versions?
- Which prompts and editing conditions?
- Which detector version?
- Independent or vendor-authored evaluation?

A metric is not an intrinsic property of a detector.

WHAT DOES “99% AI” MEAN? I

A displayed number may mean different things:

- classifier score,
- calibrated class probability,
- percentage of qualifying text,
- confidence category,
- posterior probability in a forensic context.

Calibration requires:

$$P(Y = 1 \mid \hat{p} = p) = p$$

A score in $[0, 1]$ is not automatically a probability.

BASE RATES MATTER I

Even excellent detectors can produce many false accusations if the base rate is low.

$$P(\mathrm{AI} \mid +) = \frac{P(+ \mid \mathrm{AI})P(\mathrm{AI})}{P(+ \mid \mathrm{AI})P(\mathrm{AI}) + P(+ \mid H)P(H)}$$

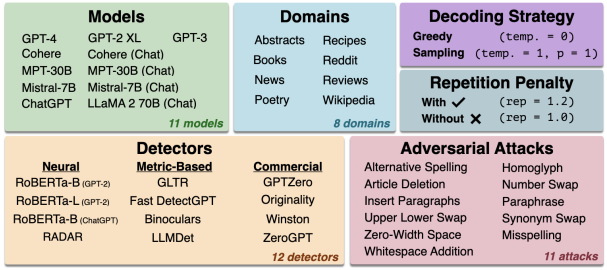
Example:

$$\mathrm{TPR} = 0.99, \quad \mathrm{FPR} = 0.01, \quad P(\mathrm{AI}) = 0.01$$

$$P(\mathrm{AI} \mid +) = 0.5$$

The same detector can imply different posteriors in different populations.

RAID: ROBUST EVALUATION I

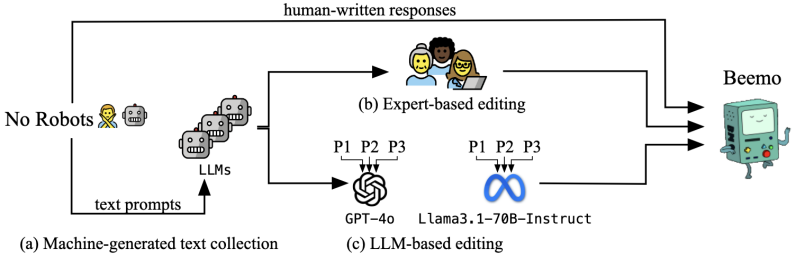


RAID evaluates detectors across: ▸ Dugan et al., 2024, Fig. 2

- generators,
- domains,
- decoding strategies,
- adversarial attacks.

Condition performance on the test distribution.

MIXED AUTHORSHIP AND EDITING I



Real texts often lie on a continuum:

human → AI-polished human → mixed → human-edited AI → AI

Binary labels become scientifically questionable.

▸ Artemova et al., 2025, Fig. 1 / Table 4

PROMPT AND STYLE ATTACKS I

Changing only the prompt can change detectability.

$$P(x \mid \text{LLM}, \text{default prompt}) \neq P(x \mid \text{LLM}, \text{style prompt})$$

Examples:

- write like Shakespeare,
- imitate a human example,
- use self-refinement to sound more human,
- avoid assistant-like phrasing.

The same generator can produce different detector signals.

▸ Zhang et al., 2024

SADASIVAN ET AL.: A THEORETICAL LIMIT I

Sadasivan et al. show that passive detection becomes impossible as machine text approaches the human distribution.

$$\mathrm{AUROC}(D) \leq \frac{1}{2} + \mathrm{TV}(P_{\mathrm{AI}}, P_{\mathrm{human}}) - \frac{1}{2}\mathrm{TV}(P_{\mathrm{AI}}, P_{\mathrm{human}})^2$$

If total variation (distributional difference) between the AI distribution P_{AI} and the human distribution P_{human} goes to zero, no passive detector can do much better than chance.

▸ Sadasivan et al., 2023

BUT: BETTER DOES NOT MEAN MORE HUMAN-LIKE I

The Sadasivan argument is valid as a mathematical statement about converging distributions.
But the empirical premise is not guaranteed:

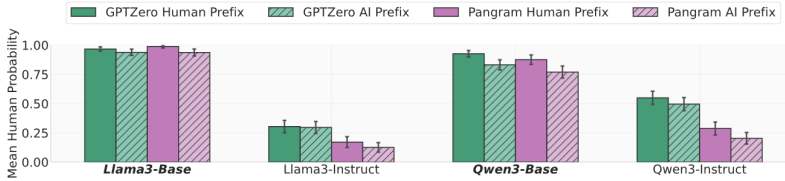
$$\text{more capable model} \not\Rightarrow P_{\mathrm{AI}} \to P_{\mathrm{human}}$$

New model versions can become:

- more instruction-following,
- more aligned to assistant conventions,
- more formulaic in product settings,
- more responsive to style constraints.

Model improvement and human-likeness are different dimensions.

BASE MODELS LOOK HUMAN TO DETECTORS I



Xu et al. find that base-model outputs can look more human to detectors than instruction-tuned outputs.

Detectors may track post-training or assistant-style artifacts, not only machine generation.

▸ Xu et al., 2026, Fig. 1

Watermarking

FROM PASSIVE DETECTION TO WATERMARKING I

Passive detection asks:

Does this text statistically look AI-generated?

Watermarking asks:

Does this text contain a signal inserted by a known generator?

Watermarking requires access to generation or deployment.

▸ Wu et al., 2025, Sec. 5.1

GREEN-RED-LIST WATERMARKING I

At each generation step:

1. use a secret key and context to split vocabulary into green/red tokens,
2. add a logit bias δ to green tokens,
3. sample normally,
4. later count green tokens.

▸ Kirchenbauer et al., 2023

Prompt	Num tokens	Z-score	p-value
…The watermark detection algorithm can be made public, enabling third parties (e.g., social media platforms) to run it themselves, or it can be kept private and run behind an API. We seek a watermark with the following properties:			
No watermark			
Extremely efficient on average term lengths and word frequencies on synthetic, microamount text (as little as 25 words) Very small and low-resource key/hash (e.g., 140 bits per key is sufficient for 99.999999999% of the Synthetic Internet	56	.31	.38
With watermark			
- minimal marginal probability for a detection attempt. - Good speech frequency and energy rate reduction. - messages indiscernible to humans. - easy for humans to verify.	36	7.4	6e-14

A SIMPLIFIED ALGORITHM I

Algorithm 1 Text Generation with Hard Red List

Input: prompt, $s^{(-N_p)} \cdots s^{(-1)}$

for $t = 0, 1, \cdots$ **do**

1. Apply the language model to prior tokens $s^{(-N_p)} \cdots s^{(t-1)}$ to get a probability vector $p^{(t)}$ over the vocabulary.
2. Compute a hash of token $s^{(t-1)}$, and use it to seed a random number generator.
3. Using this seed, randomly partition the vocabulary into a "green list" G and a "red list" R of equal size.
4. Sample $s^{(t)}$ from G , never generating any token in the red list.

end for

THE ACTUAL ALGORITHM I

Algorithm 2 Text Generation with Soft Red List

Input: prompt, $s^{(-N_p)} \cdots s^{(-1)}$
green list size, $\gamma \in (0, 1)$
hardness parameter, $\delta > 0$

for $t = 0, 1, \cdots$ **do**

1. Apply the language model to prior tokens $s^{(-N_p)} \cdots s^{(t-1)}$ to get a logit vector $l^{(t)}$ over the vocabulary.

2. Compute a hash of token $s^{(t-1)}$, and use it to seed a random number generator.

3. Using this random number generator, randomly partition the vocabulary into a "green list" G of size $\gamma|V|$, and a "red list" R of size $(1-\gamma)|V|$.

4. Add δ to each green list logit. Apply the softmax operator to these modified logits to get a probability distribution over the vocabulary.

$$\hat{p}_k^{(t)} = \begin{cases} \frac{\exp(l_k^{(t)}+\delta)}{\sum_{i\in R}\exp(l_i^{(t)})+\sum_{i\in G}\exp(l_i^{(t)}+\delta)}, & k \in G \\ \frac{\exp(l_k^{(t)})}{\sum_{i\in R}\exp(l_i^{(t)})+\sum_{i\in G}\exp(l_i^{(t)}+\delta)}, & k \in R. \end{cases}$$

5. Sample the next token, $s^{(t)}$, using the watermarked distribution $\hat{p}^{(t)}$.

end for

THE WATERMARK TEST STATISTIC I

Let:

- T: number of tested tokens,
- G: observed number of green tokens,
- γ: expected green-token fraction under the null.

Under the null:

$$G \sim \mathrm{Binomial}(T, \gamma)$$

Test statistic:

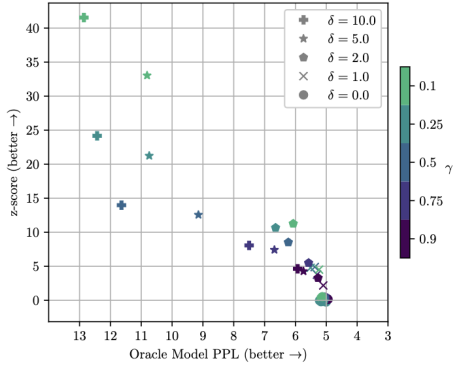
$$z = \frac{G - \gamma T}{\sqrt{T\gamma(1-\gamma)}}$$

The evidence accumulates with text length.

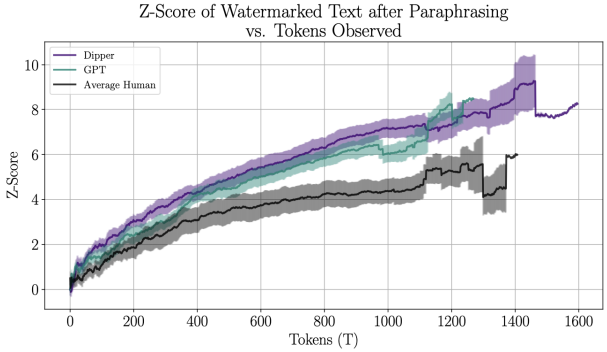
▸ Kirchenbauer et al., 2023

DETECTABILITY VS QUALITY I

Increasing the watermark strength δ improves detectability but can distort text quality. ▸ Kirchenbauer et al., 2023, Fig. 2



DO WATERMARKS SURVIVE EDITS? I



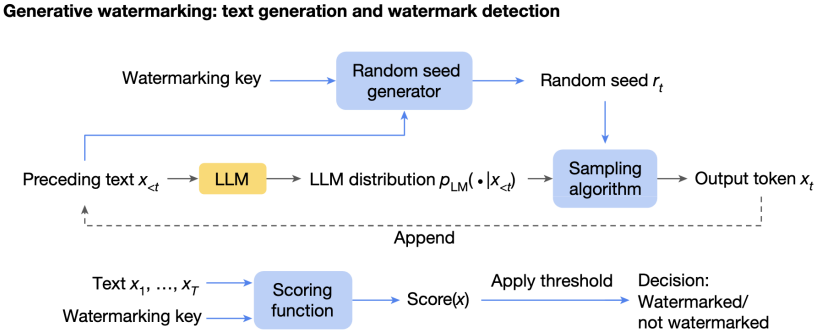
Watermarks can degrade under paraphrasing, rewriting, cropping, and mixing.

But detection can still succeed if enough signal remains.

▸ Kirchenbauer et al., 2024, Fig. 4

SYNTHID-TEXT I

SynthID-Text is a keyed generation-time watermark deployed by Google. ▸ Dathathri et al., 2024, Nature



High-level idea:

context+secret key $\longrightarrow$ modified sampling $\longrightarrow$ detectable statistical signal

GREEN LISTS VS. TOURNAMENT SAMPLING I

Both methods derive a pseudo-random signal from the **secret key + recent context**.

Kirchenbauer et al.

At each position, split the vocabulary into **green** and **red** tokens.

$$\hat{\ell}_v = \begin{cases} \ell_v + \delta, & v \in G_t \\ \ell_v, & v \notin G_t \end{cases}$$

ℓ_v: original LM logit for token v

- boost green-token probabilities
- detect unusually many green tokens

SynthID-Text

Generate a keyed pseudo-random signal

$$r_t = f_r(x_{<t}, k)$$

from context $x_{<t}$ and secret key k.

- sample plausible tokens from the LM
- rank them using keyed scores
- tournament winner becomes the next token
- detect unusually high keyed scores

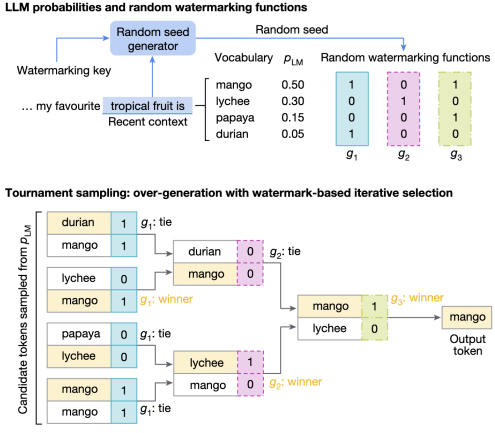
SYNTHID: TOURNAMENT SAMPLING I

At each position:

1. sample several plausible next tokens
2. score them using the secret key and context
3. let candidates compete in tournament rounds
4. output the winner

The watermark changes **which plausible token is selected**, not which tokens are possible.

▸ Dathathri et al., 2024



SYNTHID: DETECTION I

Given the text and secret key, the detector reconstructs the same pseudo-random signal: $r_t = f_r(x_{<t}, k)$
For each token x_t, it computes keyed scores $g_\ell(x_t, r_t)$.

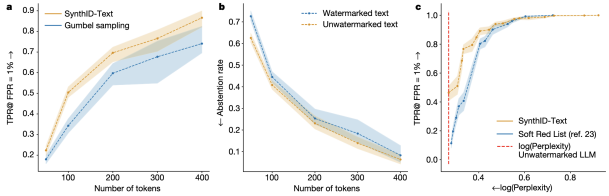
$$S(x) = \frac{1}{mT} \sum_{t=1}^{T} \sum_{\ell=1}^{m} g_\ell(x_t, r_t)$$

g_ℓ: keyed score in round ℓ; m: number of tournament rounds.
Watermarked text should have a systematically higher $S(x)$ than ordinary text: $E[S(x) \mid \text{watermarked}] > E[S(x) \mid \text{unwatermarked}]$

- evidence accumulates over many tokens
- detection requires the secret key
- the original generating LLM is not required

▸ Dathathri et al., 2024

DETECTABILITY, ABSTENTION, AND QUALITY I



Key experimental observations:

- longer texts provide stronger evidence: $T \uparrow \quad \Rightarrow \quad \mathrm{TPR} \uparrow$
- selective prediction allows abstention when evidence is weak

$$S(x) < \tau_0 : \text{unwatermarked} \qquad S(x) > \tau_1 : \text{watermarked}$$

- better detection rate than Soft Red List for high-likelihood generations

▸ Dathathri et al., 2024, Fig. 3

QUALITY, PRODUCTION DEPLOYMENT, AND LIMITS I

Quality Aspect	Prefer watm.	Prefer unwatm.	Tie: high quality	Tie: low quality	A:B [%] excl. ties	Trinom. test p
Gramm./Coher.	369	328	2089	214	52.9 [49.2, 56.7]	0.12
Relevance	358	325	1931	386	52.4 [48.6, 56.1]	0.21
Correctness	450	460	1677	413	49.5 [46.3, 52.8]	0.75
Helpfulness	793	751	1011	445	51.4 [48.8, 53.9]	0.29
Overall quality	859	805	887	449	51.6 [49.2, 54.1]	0.19

Reported results:

- large-scale live experiment on Gemini responses
- no statistically significant degradation in thumbs-up/down feedback
- human preference study found no significant preference between watermarked and unwatermarked outputs
- low reported latency overhead for tournament sampling

Important limitations:

- short text gives little statistical evidence

QUALITY, PRODUCTION DEPLOYMENT, AND LIMITS II

- low-entropy generation gives little room to encode a watermark
- paraphrasing/editing can weaken the signal
- watermark detection proves only evidence for a **specific keyed provenance signal**

▸ Dathathri et al., 2024, Extended Data Table 1

TAKEAWAYS I

1. Passive detectors can be strong in-domain, but their scores depend on distribution, prompt, model, domain, and editing.
2. Stylometric and likelihood signals are useful, but they are not immutable fingerprints of AI authorship.
3. Commercial detectors increasingly rely on supervised learning, current training data, hard negatives, and mixed-authorship labels.
4. Watermarking changes the problem from post-hoc inference to keyed provenance testing.